

Vector Spaces

Coordinate Vectors - Subspaces

Lecturer **Đặng Thị Kim Nhung**



Faculty of Applied Mathematics, Department of Applied Sciences, Ho Chi Minh City University of Technology,
268 Ly Thuong Kiet Street, District 10, Ho Chi Minh City.



E-mail: nhungdtk@hcmut.edu.vn

Coordinate Vectors

1 Coordinate Vectors

2 Subspaces

3 Null spaces

4 Change of Basis



Theorem

Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis for a vector space V . Then for each $\mathbf{x} \in V$, there exists a unique set of scalars $c_1, \dots, c_n$ such that

$$\mathbf{x} = c_1 \mathbf{b}_1 + \dots + c_n \mathbf{b}_n$$



Definition

Suppose $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a basis for a vector space V and $\mathbf{x} \in V$. The **coordinates of $\mathbf{x}$ relative to the basis $\mathcal{B}$** (or the **$\mathcal{B}$ -coordinates of $\mathbf{x}$**) are the weights $\{c_1, \dots, c_n\}$ such that $\mathbf{x} = c_1\mathbf{b}_1 + \dots + c_n\mathbf{b}_n$.

If $c_1, \dots, c_n$ are the $\mathcal{B}$ -coordinates of $\mathbf{x}$, then the vector in $\mathbb{R}^n$

$$[\mathbf{x}]_{\mathcal{B}} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}$$

is the **coordinate vector of $\mathbf{x}$ (relative to $\mathcal{B}$)**, or the $\mathcal{B}$ -coordinate vector of $\mathbf{x}$. The mapping $\mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$ is the coordinate mapping (determined by $\mathcal{B}$).



Example

Let $\mathcal{B} = \{x^2 + x + 1; x^2 + 2x + 1; x^2 + x + 2\}$ be basis of $\mathbb{R}_2[x]$. Find the vector $p(x)$, whose coordinate vector is

$$[p(x)]_{\mathcal{B}} = \begin{pmatrix} 3 \\ -5 \\ 2 \end{pmatrix}.$$



Example

Let $\mathcal{B} = \{(1, 1, 1); (1, 0, 1); (1, 1, 0)\}$ be a basis of $\mathbb{R}_3$. Find the coordinates of $\mathbf{x}$ w.r.t the basis $\mathcal{B}$.



Example

Let $\mathcal{B} = \{x^2 + x + 1; x + 1; 2x + 1\}$ be a basis of $\mathbb{R}_2$. Find the coordinates of vector $f(x) = 3x^2 + 4x - 1$ w.r.t the basis $\mathcal{B}$.



Subspaces

- 1 Coordinate Vectors
- 2 Subspaces**
- 3 Null spaces
- 4 Change of Basis



Definition

A **subspace** of a vector space V is a subset H of V that has three properties:

- a** The zero vector of V is in H .
- b** H is closed under vector addition. That is, for each u and v in H , the sum $u + v$ is in H .
- c** H is closed under multiplication by scalars. That is, for each u in H and each scalar c , the vector cu is in H .



Example

- $\mathbb{R}^2$ is not a subspace of $\mathbb{R}^3$. A plane in $\mathbb{R}^3$ not through the origin is not a subspace of $\mathbb{R}^3$.
- Given $\mathbf{v}_1, \mathbf{v}_2 \in V$, let $H = \langle \mathbf{v}_1, \mathbf{v}_2 \rangle$. Show that H is a subspace of V .



Example

Let $\mathbf{x} = \{1, -2, 3\}$; $M = \{(1, 1, 1); (2, 1, 0), (3, -1, 3)\}$. Is $\mathbf{x}$ in the subspace spanned by M ?



Example

For what values of h will $\mathbf{y}$ be in the subspace of $\mathbb{R}^3$ spanned by $\mathbf{u}, \mathbf{v}, \mathbf{w}$ if

$$\mathbf{u} = \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 5 \\ -4 \\ -7 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix}, \mathbf{y} = \begin{pmatrix} -4 \\ 3 \\ h \end{pmatrix}.$$



Example

Let $H = \langle a - 3b, b - a, a, b \rangle$. Show that H is a subspace of $\mathbb{R}^4$. Then find its dimension and a basis.



Example

Let $H = \{A \in M_2(\mathbb{R}) : A \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix} = \mathbf{0}\}$. Show that H is a subspace of $M_2(\mathbb{R})$. Then find its dimension and a basis.

To show that H is a subspace of $M_2(\mathbb{R})$. We check three properties: Zero matrix is in H , H is closed under addition and scalar multiplication. In deed, H is a subspace of $M_2(\mathbb{R})$.

- We have $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, so the zero matrix satisfies the defining condition of H . Thus, $\mathbf{0} \in H$.
- Let $A, B \in H$ i.e., $A \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix} = \mathbf{0}$, $B \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix} = \mathbf{0}$. Then
 $(A + B) \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix} = A \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix} + B \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix} = \mathbf{0} + \mathbf{0} = \mathbf{0}$. So $(A + B) \in H$.
- Let $k \in \mathbb{R}$, $A \in H$: then $(kA) \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix} = k \left(A \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix} \right) = k \cdot \mathbf{0} = \mathbf{0}$. So $kA \in H$.



Example (cont.)

Let $H = \{A \in M_2(\mathbb{R}) : A \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix} = \mathbf{0}\}$. Show that H is a subspace of $M_2(\mathbb{R})$. Then find its dimension and a basis.

Find the dimension and a basis. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then

$$A \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix} = \begin{pmatrix} a+2b & -a-2b \\ c+2d & -c-2d \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \Leftrightarrow \begin{cases} a+2b=0 \\ c+2d=0 \end{cases} \Leftrightarrow \begin{cases} a=-2b \\ c=-2d \end{cases}$$

i.e., every $A \in H$ has the form: $A = \begin{pmatrix} -2b & b \\ -2d & d \end{pmatrix} = b \begin{pmatrix} -2 & 1 \\ 0 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix}$. Thus

$$H = \text{Span} \left\{ \begin{pmatrix} -2 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix} \right\}, \text{ so } \dim(H) = 2, \text{ and basis}$$

$$B = \left\{ \begin{pmatrix} -2 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix} \right\}.$$



Example

Let $H = \langle (1, 1, 1); (2, 1, 1); (3, 1, 1) \rangle$. Find its dimension and a basis.



Example

Let $H = \langle x^2 + x + 1, 2x^2 + 3x - 1, x^2 + 2x - 2 \rangle$. Find its dimension and a basis.

Solution Form the coefficient matrix of the polynomials and compute its rank.

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 3 & 2 \\ 1 & -1 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & -3 & -3 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \text{rank}(A) = 2.$$

Thus $\dim(H) = 2$. A basis is $\{x^2 + x + 1, 2x^2 + 3x - 1\}$ (or $\{x^2 + x + 1, x^2 + 2x - 2\}$).



Example

Let $H = \left\{ \begin{pmatrix} a+b & a-2b \\ b & 2a \end{pmatrix} : a, b \in \mathbb{R} \right\}$. Find its dimension and a basis.

Solution Rewrite $\begin{pmatrix} a+b & a-2b \\ b & 2a \end{pmatrix} = a \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} + b \begin{pmatrix} 1 & -2 \\ 1 & 0 \end{pmatrix}$.

So $H = \text{span} \left\{ \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 1 & -2 \\ 1 & 0 \end{pmatrix} \right\}$.

Check if the two matrices are linearly independent. Suppose

$$\alpha \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} + \beta \begin{pmatrix} 1 & -2 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \Leftrightarrow \begin{pmatrix} \alpha + \beta & \alpha - 2\beta \\ \beta & 2\alpha \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \Leftrightarrow \begin{cases} \beta = 0 \\ \alpha = 0 \end{cases}.$$

Thus there are 2 independent generators, so $\dim(H) = 2$, and a basis for H is

$$\left\{ \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 1 & -2 \\ 1 & 0 \end{pmatrix} \right\}.$$



Null spaces

- 1 Coordinate Vectors
- 2 Subspaces
- 3 Null spaces**
- 4 Change of Basis



Consider the following system of homogeneous equations:

$$\begin{cases} x_1 - 3x_2 - 2x_3 = 0 \\ -5x_1 + 9x_2 + x_3 = 0 \end{cases}$$

In matrix form, this system is written as $A\mathbf{x} = \mathbf{0}$, where

$$A = \begin{pmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{pmatrix}; \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

Recall that the set of all $\mathbf{x}$ that satisfy the homogeneous system is called the **solution set** of the system.

We call the set of $\mathbf{x}$ that satisfy $A\mathbf{x} = \mathbf{0}$ the **null space** of the matrix A .



Definition

The **null space** of an $m \times n$ matrix A , written as $Nul(A)$, is the set of all solutions of the homogeneous equation $A\mathbf{x} = \mathbf{0}$. In set notation,

$$Nul(A) = \{\mathbf{x} : \mathbf{x} \in \mathbb{R}^n \text{ and } A\mathbf{x} = \mathbf{0}\}$$

Theorem

The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$.



Example

Let $A = \begin{pmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{pmatrix}$ and $\mathbf{u} = \begin{pmatrix} 5 \\ 3 \\ -2 \end{pmatrix}$. Determine if $\mathbf{u}$ belongs to the null space of A .

We test if $\mathbf{u}$ satisfies $A\mathbf{u} = \mathbf{0}$.

$$A\mathbf{u} = \begin{pmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 5 - 9 + 4 \\ -25 + 27 - 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Thus $\mathbf{u} \in \text{Nul}(A)$.



Example

Find the basis and dimension of the null space of the following matrix

$$A = \begin{pmatrix} 1 & 1 & 2 & 1 & 1 \\ 2 & 3 & 4 & 3 & 1 \\ 3 & 2 & 5 & 4 & 3 \\ 1 & -1 & 1 & 1 & 2 \end{pmatrix}$$



Example

Let

$$A = \begin{pmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{pmatrix}$$

- a. If the column space of A is a subspace of $\mathbb{R}^k$, what is k ?
- b. If the null space of A is a subspace of $\mathbb{R}^k$, what is k ?



Example

Let

$$A = \begin{pmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{pmatrix}$$

- a. If the column space of A is a subspace of $\mathbb{R}^k$, what is k ?
- b. If the null space of A is a subspace of $\mathbb{R}^k$, what is k ?

Solution

- a. The columns of A each have three entries, so $\text{Col}(A)$ is a subspace of $\mathbb{R}^k$, where $k = 3$.
- b. A vector $\mathbf{x}$ such that $A\mathbf{x}$ is defined must have four entries, so $\text{Nul}(A)$ is a subspace of $\mathbb{R}^k$, where $k = 4$.



Example

Find a nonzero vector in $Col(A)$ and a nonzero vector in $Nul(A)$.

$$A = \begin{pmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{pmatrix}$$

Any column of A is a vector in $Col(A)$, so we take $\begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$. To find a nonzero vector in $Nul(A)$, row reduce the augmented matrix $(A \quad \mathbf{0})$ and obtain

$$(A \quad \mathbf{0}) \sim \begin{pmatrix} 1 & 0 & 9 & 0 & 0 \\ 0 & 1 & -5 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

Thus if $\mathbf{x}$ satisfies $A\mathbf{x} = \mathbf{0}$, then $x_1 = -9x_3$, $x_2 = 5x_3$, $x_4 = 0$, and x_3 is free. Assigning $x_3 = 1$, we obtain a vector in $Nul(A)$, namely, $\mathbf{x} = (-9, 5, 1, 0)$.



Theorem

Let $A\mathbf{x} = \mathbf{0}$ be a homogeneous system and $\mathbf{x} \in \mathbb{R}^n$. Then

$$\dim(\text{Nul}(A)) + \dim(\text{Col}(A)) = \dim(\text{Nul}(A)) + \text{rank}(A) = n$$



Example

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$, $\mathbf{x} \in \mathbb{R}^3$.

Since $\text{row}_2 = 2 \times \text{row}_1$, there is only one linearly independent row, so $\text{rank}(A) = 1$ and thus $\dim(\text{Col}(A)) = 1$.

Solve $A\mathbf{x} = \mathbf{0} \Leftrightarrow x_1 + 2x_2 + 3x_3 = 0$. Let $x_2 = s$, $x_3 = t$. Then $x_1 = -2s - 3t$. Thus

$$\mathbf{x} = s \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix}$$

The null space has 2 basic vectors, hence $\dim(\text{Nul}(A)) = 2$. Thus we have

$$\dim(\text{Nul}(A)) + \dim(\text{Col}(A)) = 2 + 1 = 3$$

Null spaces

- 1 Coordinate Vectors
- 2 Subspaces
- 3 Null spaces
- 4 Change of Basis**



Consider the two coordinate systems.

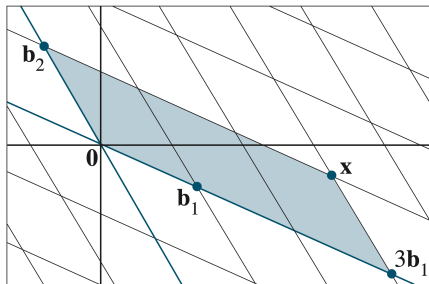


Figure: $x = 3b_1 + b_2$

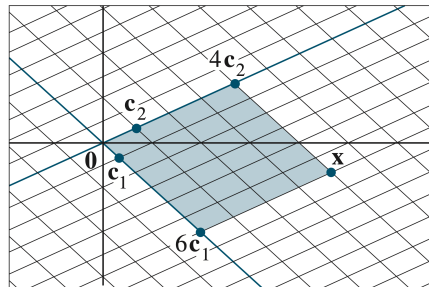


Figure: $x = 6c_1 + 4c_2$

$$[x]_B = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \text{ and } [x]_C = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$

What is the connection between the two coordinate vectors?

Example

Consider two bases $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$, $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2\}$ for a vector space V , such that $\mathbf{b}_1 = 4\mathbf{c}_1 + \mathbf{c}_2$ and $\mathbf{b}_2 = -6\mathbf{c}_1 + \mathbf{c}_2$.

Suppose $\mathbf{x} = 3\mathbf{b}_1 + \mathbf{b}_2$. That is $[\mathbf{x}]_{\mathcal{B}} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$. Find $[\mathbf{x}]_{\mathcal{C}}$.

Apply the coordinate mapping determined by $\mathcal{C}$ to $\mathbf{x} = 3\mathbf{b}_1 + \mathbf{b}_2$.

$$\begin{aligned} [\mathbf{x}]_{\mathcal{C}} &= [3\mathbf{b}_1 + \mathbf{b}_2]_{\mathcal{C}} \\ &= 3[\mathbf{b}_1]_{\mathcal{C}} + [\mathbf{b}_2]_{\mathcal{C}} \\ &= ([\mathbf{b}_1]_{\mathcal{C}} \quad [\mathbf{b}_2]_{\mathcal{C}}) \begin{pmatrix} 3 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 4 & -6 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}. \end{aligned}$$

The $\mathcal{C}$ -coordinates of $\mathbf{x}$ match those of the $\mathbf{x}$ in the figure.



Theorem

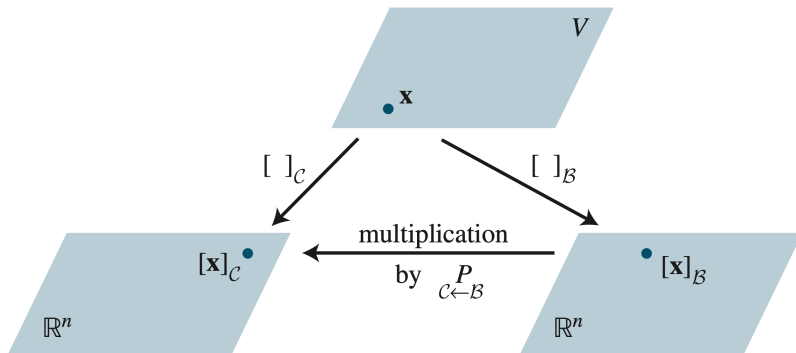
Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$, and $\mathcal{C} = \{\mathbf{c}_1, \dots, \mathbf{c}_n\}$ be bases of a vector space V . Then there is a unique $n \times n$ matrix $P_{\mathcal{C} \leftarrow \mathcal{B}}$, called the **change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$** , such that

$$[\mathbf{x}]_{\mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}} [\mathbf{x}]_{\mathcal{B}}$$

The columns of $P_{\mathcal{C} \leftarrow \mathcal{B}}$ are the $\mathcal{C}$ -coordinate vectors of the vectors in the basis $\mathcal{B}$. That is,

$$P_{\mathcal{C} \leftarrow \mathcal{B}} = ([\mathbf{b}_1]_{\mathcal{C}} \quad [\mathbf{b}_2]_{\mathcal{C}} \quad \cdots \quad [\mathbf{b}_n]_{\mathcal{C}})$$





Properties

- $[x]_C = P_{C \leftarrow B} [x]_B$ (with $P_{C \leftarrow B} = P_C^{-1} P_B$, where $P_B = (b_1, \dots, b_n)$, $P_C = (c_1, \dots, c_n)$)
- $P_{B \leftarrow C} = \left(P_{C \leftarrow B} \right)^{-1}$
- $P_{C \leftarrow A} = P_{C \leftarrow B} P_{B \leftarrow A}$



Example

Let $\mathbf{b}_1 = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$, $\mathbf{b}_2 = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$, $\mathbf{c}_1 = \begin{pmatrix} -7 \\ 9 \end{pmatrix}$, $\mathbf{c}_2 = \begin{pmatrix} -5 \\ 7 \end{pmatrix}$, and consider the bases for $\mathbb{R}^2$ given by $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2\}$.

- a Find the change-of-coordinates matrix from $\mathcal{C}$ to $\mathcal{B}$.
- b Find the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$.

Solution

$$\mathbf{a} \quad (\mathbf{b}_1 \quad \mathbf{b}_2 \mid \mathbf{c}_1 \quad \mathbf{c}_2) = \left(\begin{array}{cc|cc} 1 & -2 & -7 & -5 \\ -3 & 4 & 9 & 7 \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & 0 & 5 & 3 \\ 0 & 1 & 6 & 4 \end{array} \right).$$

$$\text{So } P_{\mathcal{B} \leftarrow \mathcal{C}} = \begin{pmatrix} 5 & 3 \\ 6 & 4 \end{pmatrix}.$$

$$\mathbf{b} \quad P_{\mathcal{C} \leftarrow \mathcal{B}} = \left(P_{\mathcal{B} \leftarrow \mathcal{C}} \right)^{-1} = \frac{1}{2} \begin{pmatrix} 4 & -3 \\ -6 & 5 \end{pmatrix} = \begin{pmatrix} 2 & -3/2 \\ -3 & 5/2 \end{pmatrix}.$$



Example (cont.)

Let $\mathbf{b}_1 = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$, $\mathbf{b}_2 = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$, $\mathbf{c}_1 = \begin{pmatrix} -7 \\ 9 \end{pmatrix}$, $\mathbf{c}_2 = \begin{pmatrix} -5 \\ 7 \end{pmatrix}$, and consider the bases for $\mathbb{R}^2$ given by $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2\}$.

- a Find the change-of-coordinates matrix from $\mathcal{C}$ to $\mathcal{B}$.
- b Find the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$.

Solution

- a Let $P_{\mathcal{B}} = (\mathbf{b}_1, \mathbf{b}_2)$, $P_{\mathcal{C}} = (\mathbf{c}_1, \mathbf{c}_2)$, then $P_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}} = \mathbf{x}$, $P_{\mathcal{C}}[\mathbf{x}]_{\mathcal{C}} = \mathbf{x}$ and $[\mathbf{x}]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}\mathbf{x}$. Thus $[\mathbf{x}]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}\mathbf{x} = P_{\mathcal{B}}^{-1}P_{\mathcal{C}}[\mathbf{x}]_{\mathcal{C}}$, which implies the change-of-coordinates matrix

$$P_{\mathcal{B} \leftarrow \mathcal{C}} = P_{\mathcal{B}}^{-1}P_{\mathcal{C}} = \begin{pmatrix} -2 & -1 \\ -1.5 & -0.5 \end{pmatrix} \begin{pmatrix} -7 & -5 \\ 9 & 7 \end{pmatrix} = \begin{pmatrix} 5 & 3 \\ 6 & 4 \end{pmatrix}.$$

- b $P_{\mathcal{C} \leftarrow \mathcal{B}} = \left(P_{\mathcal{B} \leftarrow \mathcal{C}} \right)^{-1}$



Remark

Denote $P_{\mathcal{B}} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$, $P_{\mathcal{C}} = (\mathbf{c}_1, \dots, \mathbf{c}_n)$ The change of basis matrix $P_{\mathcal{C} \leftarrow \mathcal{B}}$ is the unique solution of the equation $P_{\mathcal{C}} P_{\mathcal{C} \leftarrow \mathcal{B}} = P_{\mathcal{B}}$ which can be determine by

- Computing $P_{\mathcal{C} \leftarrow \mathcal{B}} = P_{\mathcal{C}}^{-1} P_{\mathcal{B}}$.
- Transforming $(P_{\mathcal{C}} | P_{\mathcal{B}})$ into $(I | P_{\mathcal{C} \leftarrow \mathcal{B}})$ (only elementary **row** operations are allowed).



Exercise

Let $\mathcal{A} = \{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ and $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ be bases for a vector space V , and suppose $\mathbf{a}_1 = 4\mathbf{b}_1 - \mathbf{b}_2$, $\mathbf{a}_2 = -\mathbf{b}_1 + \mathbf{b}_2 + \mathbf{b}_3$, and $\mathbf{a}_3 = \mathbf{b}_2 - 2\mathbf{b}_3$.

- a Find the change-of-coordinates matrix from $\mathcal{A}$ to $\mathcal{B}$.
- b Find $[\mathbf{x}]_{\mathcal{B}}$ for $\mathbf{x} = 3\mathbf{a}_1 + 4\mathbf{a}_2 + \mathbf{a}_3$.



Exercise

Let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2\}$ be bases for vector space $\mathbb{R}^2$. Find the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$ and the change-of-coordinates matrix from $\mathcal{C}$ to $\mathcal{B}$.

a $\mathbf{b}_1 = \begin{pmatrix} 7 \\ 5 \end{pmatrix}, \mathbf{b}_2 = \begin{pmatrix} -3 \\ -1 \end{pmatrix}, \mathbf{c}_1 = \begin{pmatrix} 1 \\ -5 \end{pmatrix}, \mathbf{c}_2 = \begin{pmatrix} -2 \\ 2 \end{pmatrix}.$

b $\mathbf{b}_1 = \begin{pmatrix} -6 \\ -1 \end{pmatrix}, \mathbf{b}_2 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \mathbf{c}_1 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \mathbf{c}_2 = \begin{pmatrix} 6 \\ -2 \end{pmatrix}.$



Exercise

Let $P = \begin{pmatrix} 1 & 2 & -1 \\ -3 & -5 & 0 \\ 4 & 6 & 1 \end{pmatrix}$, $\mathbf{v}_1 = \begin{pmatrix} -2 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} -8 \\ 5 \\ 2 \end{pmatrix}$, $\mathbf{v}_3 = \begin{pmatrix} -7 \\ 2 \\ 6 \end{pmatrix}$.

- a** Find a basis $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ for $\mathbb{R}^3$ such that P is the change-of-coordinates matrix from $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ to be the basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.
- b** Find a basis $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ for $\mathbb{R}^3$ such that P is the change-of-coordinates matrix from $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ to be the basis $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$.

Answer

a $\mathbf{u}_1 = \begin{pmatrix} -6 \\ -5 \\ 21 \end{pmatrix}$, $\mathbf{u}_2 = \begin{pmatrix} -6 \\ -9 \\ 32 \end{pmatrix}$, $\mathbf{u}_3 = \begin{pmatrix} -5 \\ 0 \\ 3 \end{pmatrix}$.

b $\mathbf{w}_1 = \begin{pmatrix} 28 \\ -9 \\ -3 \end{pmatrix}$, $\mathbf{w}_2 = \begin{pmatrix} 38 \\ -13 \\ 2 \end{pmatrix}$, $\mathbf{w}_3 = \begin{pmatrix} 21 \\ -7 \\ 3 \end{pmatrix}$.

